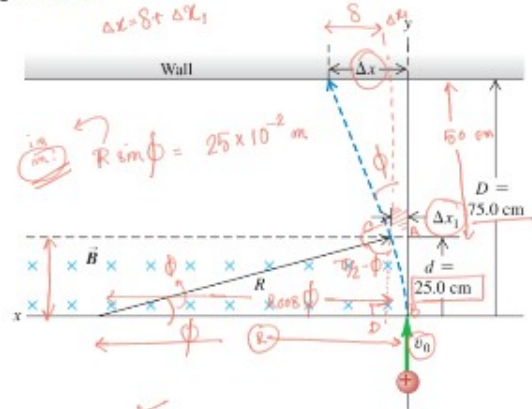


27.81 A particle with charge $2.15 \mu\text{C}$ and mass $3.20 \times 10^{-11} \text{ kg}$ is initially traveling in the $+y$ -direction with a speed $v_0 = 1.45 \times 10^5 \text{ m/s}$. It then enters a region containing a uniform magnetic field that is directed into, and perpendicular to, the page in Fig. P27.81. The magnitude of the field is 0.420 T . The region extends a distance of 25.0 cm along the initial direction of travel; 75.0 cm from the point of entry into the magnetic field region is a wall. The length of the field-free region is thus 50.0 cm . When the charged particle enters the magnetic field, it follows a curved path whose radius of curvature is R . It then leaves the magnetic field after a time t_1 , having been deflected a distance Δx_1 . The particle then travels in the field-free region and strikes the wall after undergoing a total deflection Δx_2 . (a) Determine the radius R of the curved part of the path. (b) Determine t_1 , the time the particle spends in the magnetic field. (c) Determine Δx_1 , the horizontal deflection at the point of exit from the field. (d) Determine Δx_2 , the total horizontal deflection.

Figure P27.81



Handwritten notes and equations:

- $F = q(\vec{v} \times \vec{B})$
- $\vec{F} \perp \vec{v} \Rightarrow |\vec{v}| = \text{constant}$ but direction changes.
- $\vec{F} = m\vec{a} = \frac{m v^2}{R}$
- $q v B = \frac{m v^2}{R} \Rightarrow R = \frac{m v}{q B}$
- $R = \frac{3.20 \times 10^{-11} \text{ kg} \times 1.45 \times 10^5 \text{ m/s}}{2.15 \times 10^{-6} \text{ C} \times 0.420 \text{ T}} = 5.138 \text{ m}$

Handwritten notes and equations:

- Given Data:
- $q = 2.15 \mu\text{C} = 2.15 \times 10^{-6} \text{ C}$
- $B = 0.420 \text{ T}$
- $v = 1.45 \times 10^5 \text{ m/s}$
- $m = 3.20 \times 10^{-11} \text{ kg}$

(a) $R = \frac{p}{qB} = \frac{mv}{qB} = \frac{3.20 \times 10^{-11} \times 1.45 \times 10^5}{2.15 \times 10^{-6} \times 0.420} = 5.138 \text{ m}$ (Ans)

(b) $R \sin \phi = 25 \times 10^{-2} \text{ m}$
 $\Rightarrow \sin \phi = \frac{25 \times 10^{-2}}{5.138}$
 $\Rightarrow \phi = \sin^{-1}\left(\frac{25 \times 10^{-2}}{5.138}\right)$
 $\Rightarrow \phi = 0.0486762 \text{ rad}$

$v = 1.45 \times 10^5 \text{ m/s}$
 $R = 5.138 \text{ m}$
 $\omega = \frac{v}{R}$
 $T = \frac{2\pi}{\omega}$
 $\phi \rightarrow T \times \frac{\phi}{2\pi} = t_1$
 $t_1 = T \times \frac{\phi}{2\pi} = \frac{2\pi R}{v} \times \frac{0.0486762}{2\pi}$
 $= \frac{5.138 \times 0.0486762}{1.45 \times 10^5}$
 $t_1 = 1.72 \times 10^{-6} \text{ s} = 1.72 \mu\text{s}$ (Ans)

(d) $\tan \phi = \frac{S}{R}$
 $\Rightarrow S = R \tan \phi$
 $\phi = 0.0486762 \text{ rad}$
 $\Rightarrow S = 5.138 \times \tan(0.0486762)$
 $\Rightarrow S = 2.4357 \text{ cm}$

$\Delta x = S + \Delta x_1$
 $= 2.4357 + 0.00857$
 $\Delta x = 3.04427 \text{ cm}$ (Ans)

(c) $\Delta x_1 = R - R \cos \phi$
 $= R(1 - \cos \phi)$
 $= 5.138(1 - \cos(0.0486762))$
 $\Delta x_1 = 0.00857 \text{ m}$
 $\Delta x_1 = 0.00857 \text{ cm}$ (Ans)